

The rank-three rectangular Δ -Springer ring: a q -integer-per-shape formula for $\text{Frob}_q(R_{3k,(k,k,k),3})$

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Abstract

For each $k \geq 1$, the graded Frobenius character of the Chou–Hanada Δ -Springer ring $R_{3k,(k,k,k),3}$ (*arXiv* 2509.24252) is

$$\text{Frob}_q(R_{3k,(k,k,k),3}) = \sum_{\substack{\lambda \vdash 3k \\ \ell(\lambda) \leq 3 \\ \lambda \succeq (k,k,k)}} q^{n(\lambda)} [K_{\lambda,(k,k,k)}]_q s_\lambda,$$

where $n(\lambda) = \lambda_2 + 2\lambda_3$, $K_{\lambda,(k,k,k)}$ is the Kostka number, and $[m]_q = 1 + q + \cdots + q^{m-1}$. Every q -multiplicity of a Schur function is thus a q -integer times a monomial: the sum of $K_{\lambda,(k,k,k)}$ consecutive q -powers starting at $q^{n(\lambda)}$. Combining the Chou–Hanada Cor. 2.33 SYT formula with the Lascoux–Schützenberger catabolisability characterisation of the cocharge Kostka–Foulkes polynomial reduces the theorem to the identity $\tilde{K}_{\lambda,(k,k,k)}(q) = q^{n(\lambda)} [K_{\lambda,(k,k,k)}]_q$, which we prove by a direct parametrisation of $\text{SSYT}(\lambda, (k,k,k))$ by the number of 2’s in the second row, computing the cocharge of each tableau’s reading word along the way.

1 Setup

We follow the Chou–Hanada conventions of *arXiv* 2509.24252, matching the earlier note [3] on $R_{2k,(k,k),2}$. Tableaux are drawn in the French convention (row 1 at the bottom; columns strictly increase upward).

Definition 1 (Cocharge; CH Def. 2.14). Given $S \in \text{SYT}(n)$, the cocharge tableau $\text{cc}(S)$ assigns the label 0 to the cell containing 1; and inductively, for $i = 1, \dots, n-1$, the cell containing $i+1$ receives the same label as i if $i+1$ lies in the same or a strictly lower row than i , and one more otherwise. The cocharge is $\text{cocharge}(S) := \sum \text{cc}(S)$. Extend to a word w (or an SSYT via its reading word) as usual via standardisation.

Definition 2 (Descents). For $S \in \text{SYT}(n)$, $\text{des}(S)$ is the number of indices $i \in \{1, \dots, n-1\}$ such that $i+1$ lies in a strictly higher row than i ; equivalently, $\text{des}(S) = \max \text{cc}(S)$.

Definition 3 (Catabolisability type; Blasiak, CH Alg. 1). Given $S \in \text{SYT}(n)$, form the reading word $\text{rw}(S)$ (concatenate rows top-to-bottom, each left-to-right) and the cocharge word $z = z_1 z_2 \cdots z_n$ by replacing each value v in $\text{rw}(S)$ by its cocharge label. Starting from $(x, \nu) = (z, \emptyset)$, iterate: let a be the last letter of x . If $\nu + \varepsilon_{a+1}$ is a partition (equivalently, $\nu_{a+1} < \nu_a$ where $\nu_0 = \infty$), remove a from x and add a box to row $a+1$ of ν ; otherwise remove a and prepend $a+1$. The terminal ν is $\text{ctype}(S)$.

The Chou–Hanada index set at parameters (n, λ, s) is

$$\Sigma_{n,\lambda,s} = \{(S, \mu) : S \in \text{SYT}(n), \text{ctype}(S|_k) \supseteq \lambda, \text{des}(S) < s, \mu \subseteq (n-k) \times (s - \text{des}(S) - 1)\},$$

where $k = |\lambda|$, $S|_k$ is the subtableau on entries $\{1, \dots, k\}$, and $\supseteq$ is dominance order.

Theorem 4 (Chou–Hanada, Theorem B & Cor. 2.33).

$$\text{Frob}_q(R_{n,\lambda,s}) = \sum_{(S,\mu) \in \Sigma_{n,\lambda,s}} q^{\text{cocharge}(S)+|\mu|} s_{\text{sh}(S)}.$$

For a partition $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ with $\ell(\lambda) \leq 3$ we write $n(\lambda) = \sum_i (i-1)\lambda_i = \lambda_2 + 2\lambda_3$.

2 Main theorem

Theorem 5. *For every integer $k \geq 1$,*

$$\text{Frob}_q(R_{3k,(k,k,k),3}) = \sum_{\substack{\lambda \vdash 3k \\ \ell(\lambda) \leq 3 \\ \lambda \supseteq (k,k,k)}} q^{n(\lambda)} [K_{\lambda,(k,k,k)}]_q s_\lambda,$$

where $K_{\lambda,(k,k,k)}$ is the Kostka number $\# \text{SSYT}(\lambda, (k, k, k))$.

Remark 6. At $q = 1$, Theorem 5 recovers the ungraded decomposition $h_k^3 = \sum_\lambda K_{\lambda,(k,k,k)} s_\lambda$, whose evaluation at 1 is the multinomial $(3k)!/(k!)^3 (\dim R_{3k,(k,k,k),3})$. The graded lift places the $K_{\lambda,(k,k,k)}$ copies of S^λ into *consecutive* graded degrees $n(\lambda), n(\lambda) + 1, \dots, n(\lambda) + K_{\lambda,(k,k,k)} - 1$.

Reduction to the modified Kostka–Foulkes sum

Fix $(n, \lambda_{\text{CH}}, s) = (3k, (k, k, k), 3)$. Then $|\lambda_{\text{CH}}| = 3k = n$, so $S|_k = S$ for every $S \in \text{SYT}(n)$, and $n - k = 0$ forces $\mu = \emptyset$. Substituting into Theorem 4,

$$\text{Frob}_q(R_{3k,(k,k,k),3}) = \sum_{\substack{S \in \text{SYT}(3k) \\ \text{des}(S) \leq 2 \\ \text{ctype}(S) \supseteq (k,k,k)}} q^{\text{cocharge}(S)} s_{\text{sh}(S)}. \quad (1)$$

Lemma 7 (Descent redundancy). *For $S \in \text{SYT}(n)$, $\ell(\text{ctype}(S)) \geq \text{des}(S) + 1$. Consequently, if $\text{ctype}(S) \supseteq (k, k, k)$ then $\text{des}(S) \leq 2$, so the descent constraint in (1) is redundant.*

Proof. Set $d = \text{des}(S)$ and let $a^* = d$ be the maximum cocharge label appearing in the initial word z of Blasiak’s algorithm (Definition 3). Consider one occurrence of a^* in z and track its “lineage” through the algorithm: at each step in which this particular letter is at the end of x , either it is consumed (a box is added to row $a^* + 1$ of ν) or it is replaced by $a^* + 1$ at position 0. In the latter case, its label monotonically increases; when eventually consumed, it contributes a box to row $\geq a^* + 1$. Hence $\ell(\text{ctype}(S)) \geq a^* + 1 = d + 1$.

If $\text{ctype}(S) \supseteq (k, k, k)$ then since both partitions have size $3k$, the last equality of partial sums forces $\ell(\text{ctype}(S)) \leq 3$, whence $d \leq 2$. $\square$

The next step is the identification with the modified Kostka–Foulkes polynomial. Let $\tilde{K}_{\lambda,\mu}(q) := q^{n(\mu)} K_{\lambda,\mu}(1/q)$ denote the *cocharge* Kostka–Foulkes polynomial; by Lascoux–Schützenberger, $\tilde{K}_{\lambda,\mu}(q) = \sum_{T \in \text{SSYT}(\lambda,\mu)} q^{\text{cocharge}(T)}$.

Theorem 8 (Lascoux–Schützenberger; catabolisable-SYT form; see Blasiak [2, Prop. 2.19], also CH Prop. 2.32). *For any $\lambda, \mu \vdash n$,*

$$\sum_{\substack{S \in \text{SYT}(\lambda) \\ \text{ctype}(S) \supseteq \mu}} q^{\text{cocharge}(S)} = \tilde{K}_{\lambda, \mu}(q).$$

Combining Lemma 7 with Theorem 8 applied at $\mu = (k, k, k)$:

$$\text{Frob}_q(R_{3k, (k, k, k), 3}) = \sum_{\lambda} \tilde{K}_{\lambda, (k, k, k)}(q) s_{\lambda}, \quad (2)$$

where the sum runs over $\lambda \vdash 3k$; dominance $\text{ctype}(S) \supseteq (k, k, k)$ automatically restricts $\text{sh}(S)$ to satisfy $\ell(\text{sh}(S)) \leq 3$ (via Lemma 7 and the $\text{des} \geq \ell - 1$ bound), and further restricts to shapes dominating (k, k, k) (since only such shapes admit an SSYT of weight (k, k, k)).

To finish, it remains to compute the modified Kostka–Foulkes polynomial $\tilde{K}_{\lambda, (k, k, k)}(q)$ explicitly.

Theorem 9. *Let $k \geq 1$ and let $\lambda \vdash 3k$ satisfy $\ell(\lambda) \leq 3$ and $\lambda \supseteq (k, k, k)$. Then*

$$\tilde{K}_{\lambda, (k, k, k)}(q) = q^{n(\lambda)} [K_{\lambda, (k, k, k)}]_q.$$

Theorem 9, combined with (2), proves Theorem 5. The remainder of the paper proves Theorem 9.

3 Enumerating $\text{SSYT}(\lambda, (k, k, k))$

Write $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ with $\lambda_3 \geq 0$ allowed. The constraints $\ell(\lambda) \leq 3$, $\lambda \supseteq (k, k, k)$, and $|\lambda| = 3k$ translate to $\lambda_3 \leq k \leq \lambda_1$ and $\lambda_1 + \lambda_2 \geq 2k$.

An SSYT T of shape λ and weight (k, k, k) has (English convention: row 1 at top, columns increasing downward):

- Row 1: k copies of 1, then y_1 copies of 2, then z_1 copies of 3, with $k + y_1 + z_1 = \lambda_1$;
- Row 2: y_2 copies of 2, then z_2 copies of 3, with $y_2 + z_2 = \lambda_2$ (row 2 contains no 1's by column strictness);
- Row 3: λ_3 copies of 3.

Weight (k, k, k) forces $y_1 + y_2 = k$ and $z_1 + z_2 + \lambda_3 = k$.

Lemma 10. *Let $\lambda_1, \lambda_2, \lambda_3$ satisfy the assumptions above. Then T is uniquely determined by the integer y_2 , and*

$$y_2 \in [\alpha, \beta] \quad \text{where} \quad \alpha := \max(\lambda_3, 2k - \lambda_1), \quad \beta := \min(\lambda_2, 2k - \lambda_2).$$

In particular $K_{\lambda, (k, k, k)} = \beta - \alpha + 1$.

Proof. Given y_2 , all other multiplicities are determined: $y_1 = k - y_2$, $z_2 = \lambda_2 - y_2$, $z_1 = \lambda_1 - k - (k - y_2) = \lambda_1 - 2k + y_2$.

Non-negativity. $y_1 \geq 0 \Leftrightarrow y_2 \leq k$; $z_1 \geq 0 \Leftrightarrow y_2 \geq 2k - \lambda_1$; $z_2 \geq 0 \Leftrightarrow y_2 \leq \lambda_2$; $y_2 \geq 0$ trivially.

Column strictness. At column $c \leq \lambda_3$: row 3 = 3, so row 2[c] must be 2; this needs $y_2 \geq c$, hence $y_2 \geq \lambda_3$.

At column c with $\lambda_3 < c \leq \lambda_2$: we need $\text{row } 1[c] < \text{row } 2[c]$. If $c > k$, $\text{row } 1[c] \in \{2, 3\}$; a violation occurs iff $\text{row } 2[c] = 2$ and $\text{row } 1[c] \in \{2, 3\}$, i.e., iff $c \leq y_2$ and $c > k$; this is impossible iff $y_2 \leq k$. Similarly, $\text{row } 1[c] = 3$ and $\text{row } 2[c] = 3$ collide iff $c > k + y_1$ and $c > y_2$; equivalently, all $c \in (\max(k, y_2), \lambda_2]$ satisfy $c \leq k + y_1 = 2k - y_2$, requiring $\lambda_2 \leq 2k - y_2$, i.e., $y_2 \leq 2k - \lambda_2$.

Collecting inequalities:

$$\max(\lambda_3, 2k - \lambda_1, 0) \leq y_2 \leq \min(k, \lambda_2, 2k - \lambda_2).$$

Under our assumptions, the lower bound simplifies: $\lambda_3 \geq 0$ and $2k - \lambda_1 \leq k \leq \lambda_2$ (from $\lambda_1 \geq k$ and dominance), so $\max(\lambda_3, 2k - \lambda_1, 0) = \max(\lambda_3, 2k - \lambda_1)$. The upper bound simplifies analogously. If $\lambda_2 \leq k$: $\alpha = \lambda_3$ (using $\lambda_1 + \lambda_3 \geq 2k$, from $\lambda_2 \leq k \Leftrightarrow \lambda_1 + \lambda_3 = 3k - \lambda_2 \geq 2k$) and $\beta = \lambda_2$. If $\lambda_2 \geq k$: $\alpha = 2k - \lambda_1$ and $\beta = 2k - \lambda_2$. Both cases give a nonempty interval $[\alpha, \beta]$ by the dominance hypotheses. $\square$

4 Cocharge via the reading word

Denote the SSYT of Lemma 10 by T_{y_2} . Its reading word (English convention: rows top-to-bottom, each row *right-to-left*) is

$$w = \underbrace{3^{z_1} 2^{y_1} 1^k}_{\text{row 1}} \underbrace{3^{z_2} 2^{y_2}}_{\text{row 2}} \underbrace{3^{\lambda_3}}_{\text{row 3}},$$

where exponents denote run lengths. Partition w into six segments $A B C D E F$, from left to right:

$$A = 3^{z_1}, B = 2^{y_1}, C = 1^k, D = 3^{z_2}, E = 2^{y_2}, F = 3^{\lambda_3}.$$

The k copies of 1 live entirely in C ; the k copies of 2 split as $b := y_1$ in B and $e := y_2$ in E ; the k copies of 3 split as $a := z_1$ in A , $d := z_2$ in D , and $f := \lambda_3$ in F .

The Lascoux–Schützenberger extraction

Recall $\text{cocharge}(w) = n(\mu) - \text{charge}(w)$ where $n(\mu) = 3k$ for $\mu = (k, k, k)$, and $\text{charge}(w)$ is the sum of charges of the k standard subwords extracted iteratively as follows: locate the leftmost 1; then scan cyclically to the right for the leftmost 2; then cyclically for the leftmost 3; repeat on the remainder. Each extracted subword is a permutation of $(1, 2, 3)$; its charge depends only on the relative left-to-right order of the marked positions.

For a subword whose 1 comes from C and whose 2/3 come from a specified pair of segments among $\{B, E\} \times \{A, D, F\}$, the position order of the three marks determines the charge (given the fixed segment order $A < B < C < D < E < F$):

segment of 2, segment of 3	(E, F)	(E, A)	(E, D)	(B, F)	(B, D)	(B, A)
position order of marks	$C < E < F$	$A < C < E$	$C < D < E$	$B < C < F$	$B < C < D$	$A < B < C$
subword as permutation	$(1, 2, 3)$	$(3, 1, 2)$	$(1, 3, 2)$	$(2, 1, 3)$	$(2, 1, 3)$	$(3, 2, 1)$
charge	0	1	1	2	2	3

(The charge of a permutation (u_1, u_2, u_3) is computed from the standard formula: label $c_1 = 0$, $c_{i+1} = c_i + \mathbf{1}[\text{pos}(i+1) < \text{pos}(i)]$.)

Round bookkeeping

Each round of extraction removes one 1 from C , one 2 from B or E , and one 3 from A , D , or F . The leftmost remaining 1 in round r is at the r -th position of C .

Since the 1's occupy consecutive positions strictly to the right of B , the leftmost 2 to the right cyclically is the leftmost remaining 2 in E (if any), and only when E is empty does the algorithm wrap around and select from B . Consequently:

- Rounds $1, \dots, e$ (E -rounds) mark 2 from E ;
- Rounds $e + 1, \dots, k$ (B -rounds) mark 2 from B .

The scan for a 3 proceeds *rightwards* from the marked 2.

E-round preference for 3. From a position in E , scanning right first traverses the remaining E (no 3's), then wraps to segment F , then A , then B , then C , then D . So the algorithm selects the leftmost remaining 3 preferring F , then A , then D .

B-round preference for 3. From a position in B , scanning right traverses C (no 3's), D , E (empty of 2's but not 3's—irrelevant), F ; wrap to A . So preference: D , then F , then A .

Case analysis

We compute $\text{charge}(T_{y_2})$ by counting subwords of each of the six types.

Case 1: $\lambda_2 \leq k$. Then $\alpha = \lambda_3$ and $\beta = \lambda_2$, so $y_2 \in [\lambda_3, \lambda_2]$. In particular $y_2 \geq \lambda_3 = f$, i.e., E -rounds outnumber F 's supply.

E-rounds ($e = y_2$ rounds): the first $f = \lambda_3$ take their 3 from F (fully consuming F). The next $e - f = y_2 - \lambda_3$ take their 3 from A (preferred over D). We verify these fit: $a - (y_2 - \lambda_3) = (\lambda_1 - 2k + y_2) - (y_2 - \lambda_3) = \lambda_1 + \lambda_3 - 2k = 3k - \lambda_2 - 2k = k - \lambda_2 \geq 0$ (by hypothesis), so A has enough 3's.

E-round totals: $(E, F) : f$, $(E, A) : y_2 - \lambda_3$, $(E, D) : 0$.

B-rounds ($b = k - y_2$ rounds): remaining 3's live in A ($k - \lambda_2$) and D ($d = \lambda_2 - y_2$). The preference is D first, so the first $\lambda_2 - y_2$ B -rounds take from D (exhausting it), and the remaining $b - d = k - \lambda_2$ take from A .

B-round totals: $(B, D) : \lambda_2 - y_2$, $(B, F) : 0$, $(B, A) : k - \lambda_2$.

Charge sum:

$$\begin{aligned} \text{charge}(w) &= 0 \cdot f + 1 \cdot (y_2 - \lambda_3) + 2 \cdot (\lambda_2 - y_2) + 3 \cdot (k - \lambda_2) \\ &= 3k - y_2 - \lambda_2 - \lambda_3. \end{aligned}$$

So $\text{cocharge}(T_{y_2}) = 3k - \text{charge}(w) = y_2 + \lambda_2 + \lambda_3 = (y_2 - \lambda_3) + (\lambda_2 + 2\lambda_3) = (y_2 - \alpha) + n(\lambda)$.

Case 2: $\lambda_2 \geq k$. Then $\alpha = 2k - \lambda_1$ and $\beta = 2k - \lambda_2$, so $y_2 \in [2k - \lambda_1, 2k - \lambda_2]$. In particular $y_2 \leq 2k - \lambda_2 \leq k$ (using $\lambda_2 \geq k$), so $b = k - y_2 \geq 0$. Also $y_2 \geq \lambda_3$ (as before).

E-rounds ($e = y_2$ rounds): the first $f = \lambda_3$ take their 3 from F (fully consuming F). The next $\min(e - f, a) = a$ take their 3 from A ; we check $e - f - a = y_2 - \lambda_3 - (\lambda_1 - 2k + y_2) = 2k - \lambda_1 - \lambda_3 = \lambda_2 - k \geq 0$, so this is tight. The remaining $\lambda_2 - k$ E -rounds take from D .

E-round totals: $(E, F) : f = \lambda_3$, $(E, A) : a = \lambda_1 - 2k + y_2$, $(E, D) : \lambda_2 - k$.

B-rounds ($b = k - y_2$ rounds): F and A are exhausted; only D remains, with $d - (\lambda_2 - k) = k - y_2$ copies. So all $b = k - y_2$ B -rounds pick from D .

B-round totals: $(B, D) : k - y_2$, $(B, F) : 0$, $(B, A) : 0$.

Charge sum:

$$\begin{aligned}\text{charge}(w) &= 0 \cdot \lambda_3 + 1 \cdot (\lambda_1 - 2k + y_2) + 1 \cdot (\lambda_2 - k) + 2 \cdot (k - y_2) \\ &= \lambda_1 + \lambda_2 - k - y_2.\end{aligned}$$

So $\text{cocharge}(T_{y_2}) = 3k - (\lambda_1 + \lambda_2 - k - y_2) = 4k - \lambda_1 - \lambda_2 + y_2$. Using $\lambda_1 + \lambda_2 = 3k - \lambda_3$:

$$\text{cocharge}(T_{y_2}) = 4k - (3k - \lambda_3) + y_2 = k + \lambda_3 + y_2 = (y_2 - (2k - \lambda_1)) + (\lambda_2 + 2\lambda_3) = (y_2 - \alpha) + n(\lambda),$$

using $\lambda_2 = 3k - \lambda_1 - \lambda_3$ in the last step.

Both cases yield the same conclusion:

Proposition 11. *For each valid $y_2 \in [\alpha, \beta]$, $\text{cocharge}(T_{y_2}) = n(\lambda) + (y_2 - \alpha)$.*

Assembling

Proof of Theorem 9. By Proposition 11, as y_2 ranges over $[\alpha, \beta]$, the cocharge takes each value in $\{n(\lambda), n(\lambda) + 1, \dots, n(\lambda) + (\beta - \alpha)\}$ exactly once. Hence

$$\tilde{K}_{\lambda, (k, k, k)}(q) = \sum_{y_2=\alpha}^{\beta} q^{n(\lambda)+(y_2-\alpha)} = q^{n(\lambda)}[\beta - \alpha + 1]_q = q^{n(\lambda)}[K_{\lambda, (k, k, k)}]_q. \quad \square$$

Proof of Theorem 5. Combine Theorem 9 with (2). $\square$

5 Verification

The formula was verified by exhaustive Python enumeration of both the Chou–Hanada higher-Specht index set (Theorem 4) and the Kostka–Foulkes cocharge sum on SSYT for $k = 1, 2, 3$ (corresponding to $n = 3, 6, 9$). All coefficients match term-by-term across all 22 non-zero shape summands, and ungraded dimensions $\dim R_{3k, (k, k, k), 3} = (3k)!/(k!)^3 = 6, 90, 1680$ are reproduced. Probes at

`~/projects/probes/2026-08-08-chou-hanada-r-three/`

(files `probe_r3.py`, `kostka_foulkes.py`, `compare_kf.py`) reuse the calibrated CH probe from the earlier $r = 2$ note and cross-check against the cocharge Kostka–Foulkes polynomial computed via Lascoux–Schützenberger’s standard-subword algorithm.

6 Discussion

The r -dependence: rank three is still special

At $r = 2$ (the earlier note [3]) every Schur coefficient was a single monomial, q^j : the descent-bound argument forced two-row shapes and one SYT per shape. At $r = 3$ the descent bound only pins shapes to three rows, and multiple SSYT of a given shape and weight (k, k, k) appear. What survives, and is genuinely new here, is that *the cocharges of these SSYT still form a contiguous run* $\{n(\lambda), n(\lambda) + 1, \dots\}$, one increment per parametric y_2 -step, producing a q -integer coefficient. This is a special feature of three-row shapes with three-row weight (k^3) ; it fails at $r = 4$ (where, e.g., $f^{(2,2)}(q) = q^2 + q^4$ has a gap).

Ring-level interpretation

The identity $\text{Frob}_q(R_{3k,(k,k,k),3}) = H'_{(k,k,k)}(x;q)$ (via (2)) says $R_{3k,(k,k,k),3}$ shares the graded Frobenius character of the DeConcini–Procesi ring $\mathcal{R}_{(k,k,k)}$ for the nilpotent orbit of Jordan type (k, k, k) . This is not surprising given that Chou–Hanada’s Δ -Springer construction specialises to Springer / Garsia–Procesi at $n = |\lambda|$, $s = \ell(\lambda)$; the content of Theorem 5 is really the closed form for the modified Kostka–Foulkes polynomial $\tilde{K}_{\lambda,(k^3)}(q)$, together with the observation that the q -multiplicities of Schur functions are single q -integers times monomials — a feature intrinsic to the three-row rectangular case.

Position vs. the composite- d project

The composite- d regular-element theorem lives on the coinvariant-scale module $H_n = R_{n,(1^n),n}$ and studies character-level Frobenius specialisations at cyclotomic points; the present result computes a graded Frobenius on the multinomial-scale parabolic-quotient $R_{3k,(k,k,k),3}$ and is orthogonal to the regular-element story, in the same way the earlier $r = 2$ note was. Together the two notes cover the first two rectangular ranks of the Chou–Hanada modules at their diagonal $s = r$ parameters.

References

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- [2] J. Blasiak, *Cyclage, catabolism, and the affine Hecke algebra*, arXiv:1208.6371 (2012); esp. Prop. 2.19.
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